

Fake Sorting

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

You are given an array a of n integers.

You may perform the following operation any number of times:

- Choose two positions i and j such that $a_i = a_j$, and swap a_i with a_j .

Determine whether it **is possible to make the array sorted** in non-decreasing order using any (possibly zero) number of operations.

Input

The first line contains an integer n ($1 \leq n \leq 10$) the number of elements in the array.

The second line contains n integers a_i ($1 \leq a_i \leq 10^{18}$) the array values.

Output

Print YES if it is possible to make the array sorted in non-decreasing order using the given operation. Otherwise, print NO.

Examples

standard input	standard output
6 1 2 3 4 2 3	NO
4 1 2 4 7	YES